

Game — Debug the Module

A short exercise in using Python's interactive debugger (`pdb`) to find a bug that lives in an **imported module**, not your main script.

The Bug

`main.py` imports `calculate_total` from `shop.py`, but the result is wrong:

```
# main.py
from shop import calculate_total

cart = [("apple", 2, 1.50),
        ("bread", 1, 3.00),
        ("milk", 3, 2.25)]

print(calculate_total(cart))

# Expected: 12.75 -> Got: 6.75
```

The bug is somewhere inside `shop.py`:

```
# shop.py
def calculate_total(cart):
    total = 0
    for name, qty, price in cart:
        total += price
    return total
# Bug is somewhere in here!
```

Your Mission

1. `main.py` imports `calculate_total` from `shop.py` — but the result is wrong.
 2. Open `shop.py` and add `breakpoint()` as the **first line inside the for loop**.
 3. Run `python main.py` from the terminal. Execution pauses inside `shop.py`.
 4. At the (Pdb) prompt, type `p name`, `p qty`, and `p price` for each item.
 5. Type `n` to step forward and watch how `total` changes — is `qty` being used?
 6. Find the missing piece, fix the line, remove `breakpoint()`, and re-run!
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Cross-File Tip

`pdb` pauses wherever `breakpoint()` is — **even if that’s inside an imported module, not your main script**. You don’t need to put the breakpoint in `main.py` to debug code that lives elsewhere; the debugger follows execution across file boundaries.

Files

File	Role
<code>shop.py</code>	The module containing <code>calculate_total()</code> — and the bug.
<code>main.py</code>	Entry point. Imports and calls <code>calculate_total()</code> .

Requirements

- Python 3.7+ (for built-in `breakpoint()` support)

Running

```
python main.py
```

Once you’ve added `breakpoint()` to `shop.py` as instructed above, this will drop you into an interactive (Pdb) session at that line.

Useful pdb commands for this exercise

Command	Effect
<code>p <expr></code>	Print the value of <code><expr></code> (e.g., <code>p qty</code>)
<code>n</code>	Next line (step over)
<code>c</code>	Continue execution until the next breakpoint
<code>l</code>	List source code around the current line
<code>q</code>	Quit the debugger